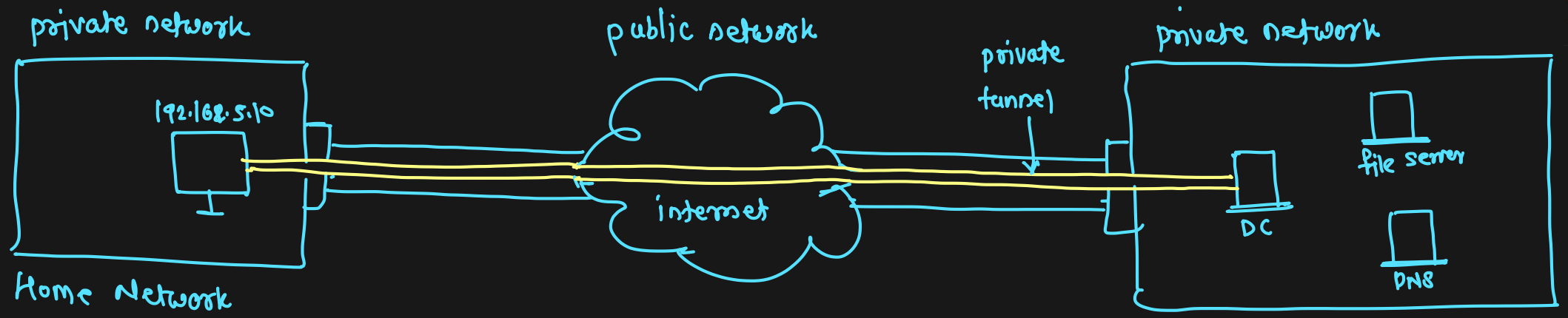
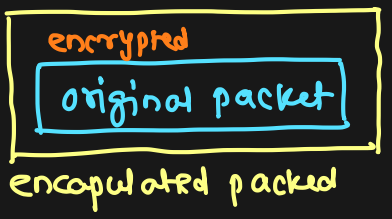


Virtual Private Network

Introduction

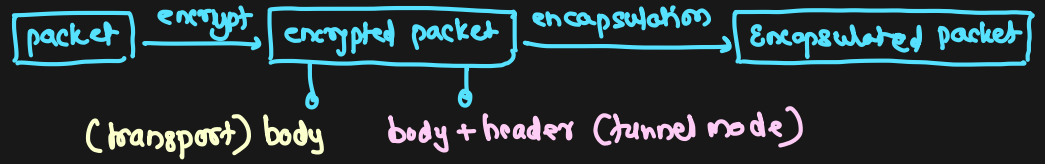
- A Virtual Private Network (VPN) is a technology that creates an **encrypted tunnel** over a public network (such as the internet), allowing remote users or branch offices to access resources as if they were directly connected to the internal network
- The core mechanism is **tunneling**: original data packets are encrypted and encapsulated inside a new packet, which travels across the public network
- Only the devices at both ends of the tunnel can decrypt and read the contents
- How doe it work?
  - VPN achieves secure communication through **encapsulation**
  - The original IP packet (including its header) is encrypted and wrapped inside a new packet
  - The outer packet uses normal routing to travel across the internet
  - The tunneling protocol determines how encapsulation and de-encapsulation are performed
  - Data confidentiality relies on symmetric encryption (such as AES-256-GCM), while key exchange typically uses Diffie-Hellman or its elliptic-curve variants



Types of VPN

- Gateway-to-Gateway
  - Purpose: Connect two networks
  - Scenario: Branch office interconnection
- Host-to-Gateway (Remote Access)
  - Purpose: Remote user accesses corporate network
  - Scenario: Remote work
- Host-to-Host
  - Purpose: Connect specific host to specific resource
  - Scenario: Point-to-point secure communication
- Third-Party Privacy VPN
  - Purpose: Protect connection on public hotspots
  - Scenario: Airport/Hotel Wi-Fi

VPN Protocol Comparison



- **IPsec (Internet Protocol Security)**
  - Recommended as the preferred protocol for enterprise VPNs; it is an open standard, so different implementations can interoperate
  - Two modes: **Transport mode** (encrypts only the payload, keeps the original IP header) and **Tunnel mode** (encrypts the entire original packet and adds a new IP header)
  - Tunnel mode is the common mode for business VPNs
- **SSTP (Secure Socket Tunneling Protocol)**
  - Based on HTTPS, **firewall-friendly**, uses TCP port 443
  - **Still fully supported** in Windows Server 2025

private key → used for decrypted
public key → used for encryption

- **IKEv2 (Internet Key Exchange version 2)**
  - Modern and stable, **mobile-device friendly**, supports keeping the connection alive when switching networks
  - Uses UDP ports 500 and 4500
- **L2TP/IPsec** ✗
  - Older protocol; **no longer accepts connections by default in new Windows Server 2025 installations**
  - Can still be enabled manually, but its security profile is insufficient — **not recommended**
- **PPTP** ✗
  - **Obsolete and insecure**; disabled by default in new Windows Server 2025 installations
  - Strongly discouraged

Network Policy Server

- **Network Policy Server (NPS)** is Microsoft's implementation of a **RADIUS server** and **RADIUS proxy**
- It is installed as part of the **Network Policy and Access Services (NPAS)** role on Windows Server
- **RADIUS** = Remote Authentication Dial-In User Service
- NPS centralizes **authentication, authorization, and accounting (AAA)** for network access
- It decides **who can connect**, **what they can access**, and **logs what they did**
- **Core Functions**
  - Authentication: Verifies the user's identity (username/password, certificate, etc.)
  - Authorization: Decides if the user is allowed to connect and under what conditions
  - Accounting: Logs connection events (who connected, when, how long, data used)

NPS Components

- **RADIUS Clients**
  - Devices that send authentication requests to NPS (e.g., RRAS, Wi-Fi APs, VPN gateways)
  - Each client must be registered in NPS with a **shared secret** or use certificate
    - → private key
    - → public key
  - In your lab: RRAS itself is the RADIUS client (localhost)
- **Network Policies**
  - Rules that determine whether a connection is allowed
  - Each policy has:
    - Conditions: Who/what must match (user group, time, IP, NAS type)
    - Constraints: Limits (idle timeout, session timeout, encryption strength)
    - Settings: What happens if matched (Grant/Deny, auth method, VLAN)
- **Connection Request Policies**
  - Decide **which NPS server** processes the request
  - Used mainly in **RADIUS proxy** scenarios (forwarding to another NPS)
  - In a single-server lab, these are rarely modified
- **Remote RADIUS Server Groups**
  - Used when NPS forwards requests to another RADIUS server
  - Not needed in your single-server lab

Authentication Methods Supported by NPS

- **PAP:** Basic, unencrypted, Avoid — sends password in clear text
- **CHAP:** Legacy Rarely used
- **MS-CHAP v2:** VPN with username/password
- **EAP-MSCHAPv2:** Needs certificate on server
- **EAP-TLS:** Certificate-based .Needs PKI (CA + client certs)
- **PEAP-TLS:** Certificate + tunnel. Enterprise Wi-Fi/VPN

NPS Policy Processing Order

- **Connection Request Policy:** picks the target NPS server
- **Network Policy:** first matching policy wins
- **If no policy matches:** connection is **rejected**
- **If Grant Access:** RRAS completes the tunnel
- **If Deny Access:** error returned to client (e.g., Error 812)

Certificate Creation

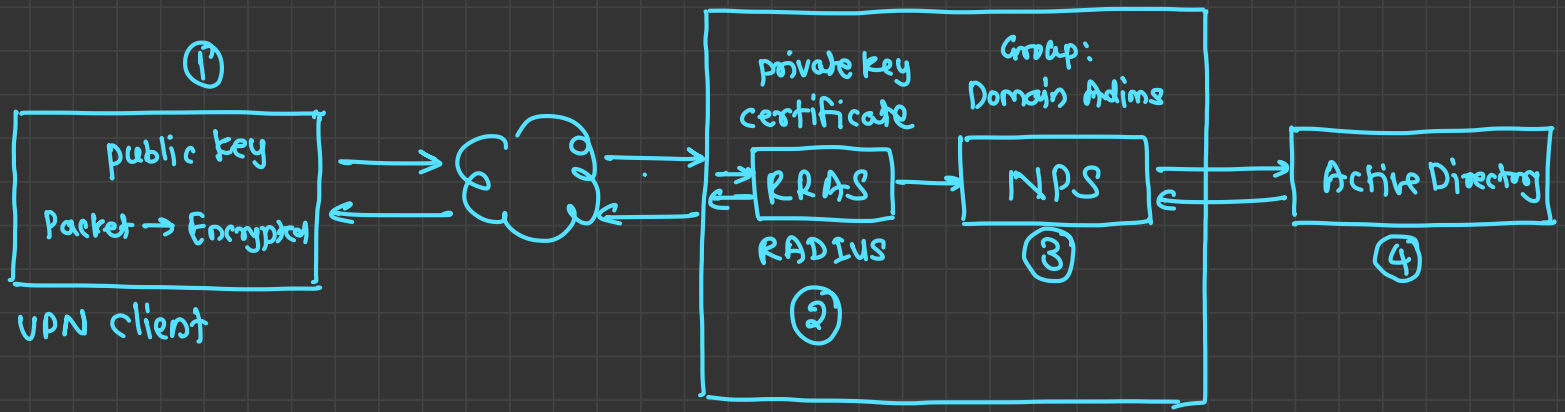


- Create Certificate

```
New-SelfSignedCertificate `
-Type Custom `
-Subject "CN=VPNLabRoot" `
-KeySpec Signature `
-KeyExportPolicy Exportable `
-KeyUsage CertSign `
-KeyUsageProperty Sign `
-KeyLength 2048 `
-HashAlgorithm sha256 `
-NotAfter (Get-Date).AddYears(5) `
-CertStoreLocation "Cert:\LocalMachine\My"
```

- Export Certificate

```
Export-Certificate `
-Cert (Get-ChildItem Cert:\LocalMachine\My\<THUMBPRINT>) `
-FilePath C:\VPNLabRoot.cer
```


①
public key
Packet → Encrypted
VPN client
private key
certificate
RRAS
RADIUS
②
Group:
Domain Admins
NPS
③
Active Directory
④

Configuring VPN

- **Prerequisites**
  - **Windows Server 2025** installed with Desktop Experience
  - Membership in the **Administrators group** or equivalent; domain-joined RAS servers require domain admin rights
  - A planned **static IP address pool** for VPN clients
- **Step 1: Install the Remote Access Role**
  - Open **Server Manager** → **Manage** → **Add Roles and Features**
  - Select **Role-based or feature-based installation**
  - Check **Remote Access** in the roles list
  - On the role services page, check **DirectAccess and VPN (RAS)**
  - Complete the installation and reboot if prompted
- **Step 2: Configure RRAS as a VPN Server**
  - Open the **Routing and Remote Access MMC** (run rrasmgmt.msc)
  - Right-click the server name and choose **Configure and Enable Routing and Remote Access**
  - Select **Custom Configuration** in the wizard
  - Check **VPN access** and finish the wizard
  - Click **OK** to close the dialog after the wizard completes

Configuring VPN

- **Step 3: Configure the IP Address Pool**
  - VPN clients need IP addresses from the server. A **static address pool** is recommended:
  - In the RRAS console, right-click the server → **Properties**
  - Select the **IPv4** tab
  - Choose **Static address pool** → **Add**
  - Enter the start and end IP addresses (ensure these are **not already assigned by DHCP**)
  - Click **OK** to save
- **Step 4: Configure VPN Protocols**
  - New RRAS installations on Windows Server 2025 **no longer accept PPTP and L2TP connections by default**
  - It is recommended to **keep only SSTP and IKEv2** and disable insecure protocols
  - In the RRAS console, right-click **Ports** → **Properties**
  - Locate **WAN Miniport (SSTP)** and click **Configure**
  - Ensure **Remote access connections (inbound only)** is checked and **Demand-dial routing connections** is cleared
  - Do the same for **WAN Miniport (IKEv2)**
  - If not needed, uncheck inbound connections for **WAN Miniport (PPTP)** and **WAN Miniport (L2TP)**
  - Restart the Routing and Remote Access service for changes to take effect

Configuring VPN

- **Step 5: Configure the Firewall**
  - Open the appropriate ports based on your chosen protocols
  - **IKEv2**: UDP **500** and **4500** inbound
  - **SSTP**: TCP **443** inbound
- **Step 6: Grant Users VPN Access**
  - Open **Active Directory Users and Computers** (domain) or **Local Users and Groups** (non-domain)
  - Edit the target user's properties
  - Switch to the **Dial-in** tab
  - Under **Network Access Permission**, select **Allow access**
  - Click **OK** to save
- **Step 7: Test the Client Connection**
  - On the client device:
    - Go to **Settings** → **Network & Internet** → **VPN** → **Add a VPN connection**
    - **VPN provider: Windows (built-in)**
    - **Connection name**: choose any name
    - **Server name or address**: enter the server's **public IP or domain name**
    - **VPN type**: select **IKEv2** or **SSTP**
    - **Type of sign-in info**: select **User name and password** and enter credentials
    - Save and click **Connect**
    - After a successful connection, verify access to internal network resources.